

Multiple Choice Questions

1.	A	6.	A
2.	D	7.	B
3.	A	8.	D
4.	A	9.	B
5.	A	10.	A

Structured Questions

1.	(a) any suitable independent variable such as - the surface - the angle of the slope - the kind of object - the size of the push <i>accept specific variations in objects, such as, 'weight' or 'mass' or 'surface area' or 'type of trainer sole' or 'type of shoe'</i> 1 (L4) (b) any suitable dependent variable such as - the distance travelled - the time to move down the ramp - the force needed to start the object moving - the angle of the ramp at which the object starts moving <i>accept 'the time to reach a given point'</i> <i>accept 'angle or height of ramp'</i> <i>accept 'speed'</i> <i>a dependent variable (DV) without an independent variable (IV) can gain credit</i> 1 (L4)
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any appropriate equipment to measure the dependent variable such as

- ruler **or** metre rule
- stopwatch **or** timer **or** light gates
- newton meter
- protractor

accept 'tape measure'

accept 'clock'

*do **not** accept a measurement strategy if a DV is not given
or is incorrect*

1 (L4)

(c) any appropriate control variable such as

- the object used
- the angle of the slope
- the surface used
- the height of the ramp
- the length of the ramp

accept 'distance travelled'

*only give credit for a control variable which does
not conflict with the suggested investigation*

1 (L4)

[4]

2.	(a) • X-axis: time (minutes) Y-axis: temperature (°C) 1 (L7) • appropriate scales such as 2 cm represents 10°C, 2 cm represents 1 minute 1 (L7) • 11 points plotted to within half a small square 1 (L7) • a smooth curve of best fit (not dot-to-dot) pupils can gain credit for correct responses to other parts if the axes are wrongly labelled both answers are required for the mark units are required for the mark at least one zero must be included at the origin 1 (L7) (b) an answer consistent with the drawn curve accept a tolerance of one small square 1 (L6) [5]															
3.	<table><tr><td>(a)(i)</td><td>46 ;</td><td>1</td></tr><tr><td>(a)(ii)</td><td>male and the presence of XY chromosomes ;</td><td>1</td></tr><tr><td>(a)(iii)</td><td>X inherited from, mother / egg / female gamete ; Y inherited from, father / sperm / male gamete ;</td><td>2</td></tr><tr><td>(b)</td><td><u>DNA</u> ; <u>genes</u> ;</td><td>2</td></tr><tr><td>(c)</td><td>joining of nuclei ; (of) sperm and egg / male and female gamete / sex cells ;</td><td>2</td></tr></table>	(a)(i)	46 ;	1	(a)(ii)	male and the presence of XY chromosomes ;	1	(a)(iii)	X inherited from, mother / egg / female gamete ; Y inherited from, father / sperm / male gamete ;	2	(b)	<u>DNA</u> ; <u>genes</u> ;	2	(c)	joining of nuclei ; (of) sperm and egg / male and female gamete / sex cells ;	2
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5.

(a) (i) any **one** from

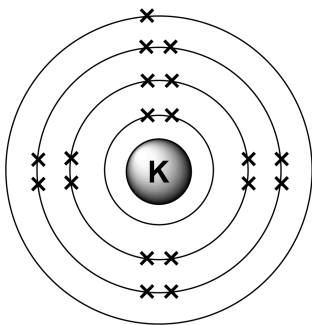
- they have inherited different characteristics
accept 'selective breeding'
- they have different genes
*accept 'they have different DNA **or** alleles'*

1

(ii) any **two** from

- amount of food
accept 'amount of grass'
- type of food
- amount of water
- levels of disease **or** quality of health care
*accept 'disease' **or** 'health care'*
- levels of stress
- temperature
accept 'climate'
*accept 'amount of exercise' **or** 'amount of space'*
accept 'their hormone levels'

2

	(b) (i) selective breeding <i>accept 'artificial selection' or 'cross breeding'</i> <i>do not accept 'inter-breeding'</i> (ii) any one from - resistance to disease <i>do not accept 'good health'</i> - fast growth rate - high fertility <i>accept 'fertile'</i> - long life - good skin for making leather - docile temperament - hardiness or resistance to weather <i>accept 'prize-winning appearance'</i> <i>accept 'females' or 'heifers'</i> <i>accept 'polled' or 'no horns'</i>	1					
		1	[5]				
6.	<table><tr><td>(a)(i)</td><td>2.8.8.1;</td></tr><tr><td>(a)(ii)</td><td>number of outer shell electrons is the same as group number;</td></tr></table> 19 electrons shown ; (2).8.8.1 ; 	(a)(i)	2.8.8.1;	(a)(ii)	number of outer shell electrons is the same as group number;		
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